sPHENIX Self-Paced Course — Module 4

Tracking detectors: MVTX, INTT, TPC and TPOT

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Self-contained study text built on the Summer-2026 sPHENIX School, public sPHENIX papers and standard detector textbooks

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How to use this module

This module is self-contained. It explains how charged particles are detected in silicon and in gas, and how the four sPHENIX tracking subsystems are built, read out and used. You do not need slides, recordings or papers to follow it. The school talks listed below are the *sources*; they are cited as [S##] so that a statement can be traced. Track reconstruction (seeding, fitting, alignment, TPC distortion corrections) is the subject of Module 8; here we concentrate on the detectors.

Learning objectives

After this module you should be able to:

O1. Explain energy loss of charged particles (Bethe–Bloch, MIP, Landau fluctuations) and multiple Coulomb scattering, and estimate the signal in silicon and in gas.

O2. Explain charge collection in a reverse-biased silicon junction, and contrast strips, hybrid pixels and monolithic active pixel sensors (MAPS).

O3. Describe the MVTX (ALPIDE MAPS, three layers) and its strobe-based readout, and explain what it contributes to vertex and DCA resolution.

O4. Describe the INTT (two barrel layers of silicon strips) and explain why it is the detector that ties tracks to a single bunch crossing.

O5. Explain drift, diffusion and gas amplification, and describe the sPHENIX TPC (GEM readout, zigzag pads, continuous readout, ion back-flow, space charge, laser calibration).

O6. Describe TPOT (Micromegas) and its role as a distortion-free reference outside the TPC.

O7. Compare integration times (MVTX strobe, INTT crossing, TPC drift) and explain the resulting pile-up and timing logic.

O8. Relate detector design to momentum and DCA resolution with back-of-the-envelope formulas.

Prerequisites and time

Modules 1 and 2. Module 3 is helpful but not required. Estimated time: 9–11 hours (about 6 hours reading, 2 hours worked examples, 1–3 hours optional exercises).

Sources (sPHENIX School 2026, BNL, July 2026)

- [S10] C. Dean, “MVTX Overview” — silicon physics, MAPS, ALPIDE readout, resolution, vertexing, alignment.
- [S11] A. Enokizono, “INTT Overview” — INTT layout, sensors, readout, timing.
- [S12] T. Hemmick, “TPC and TPOT Overview” — gas detectors, TPC mechanics and physics, space charge, laser calibration, TPOT.
- [S28] T. Kikuchi, “INTT hit detection efficiency study” (flash talk) — how a hit efficiency is measured with tracks.

Public readings (optional)

INTT ladder paper [R1] and beam-test paper [R2]; the INTT detector paper [R3]; sPHENIX TPC design proceedings [R4, R5]; the sPHENIX Au+Au multiplicity paper, which describes the INTT and MVTX as

used in a physics result [R6]; the PDG reviews of passage of particles through matter and of particle detectors [R7].

Notation

p_T in GeV/c, B in tesla, lengths in metres unless stated. "DCA" = distance of closest approach of a track to the vertex (also called impact parameter). "BCO" = beam-clock counter, i.e. a RHIC bunch-crossing number. One RHIC crossing lasts about 106 ns. X_0 = radiation length; x/X_0 = material thickness in units of X_0.

1 The sPHENIX tracking system at a glance

Goal. Know which detector sits where, what it measures, and why the system needs all four.

1.1 What a tracker must do

A tracking system has three main tasks [S10]:

1. record the trajectories of long-lived charged particles (pions, kaons, protons, electrons, muons);
2. find the collision point (primary vertex) and any displaced decay points (secondary vertices);
3. measure momentum and charge from the curvature in the magnetic field.

Secondary tasks are timing (which bunch crossing did a track come from?) and particle identification (from specific energy loss $\mathrm{d}E/\mathrm{d}x$ in gas) [S10].

1.2 The four subsystems

Detector	Technology	Radius	Main role
MVTX	3 layers of MAPS pixels (ALPIDE)	≈ 2.5–4 cm	vertex and DCA precision
INTT	2 layers (4 sub-layers) of Si strips	$\approx$ 7–10 cm	bunch-crossing timing; links MVTX and TPC
TPC	gas TPC, quad-GEM, zigzag pads	≈ 20–78 cm	momentum, pattern recognition, $\mathrm{d}E/\mathrm{d}x$
TPOT	Micromegas, limited ϕ coverage	just outside the TPC	distortion-free reference for TPC calibration

All of them cover roughly $|\eta| < 1.1$ (TPOT only partially in ϕ) inside the 1.4 T solenoid [S10, S11, S12, R4, R6].

1.3 Why four detectors?

Each technology is good at something different:

- **MVTX** is very close to the beam and very thin, so it gives the best pointing to the vertex. It is slow (microsecond integration).
- **INTT** is fast: it records which 106 ns crossing a hit belongs to. It is coarse along z.
- **TPC** gives many points (48 pad rows) over a long lever arm, which is what measures momentum. It is very slow (about 13 μs maximum drift) and its points are displaced by drift distortions.
- **TPOT** measures points just outside the TPC with negligible drift, which lets the TPC distortions be measured from outside [S12].

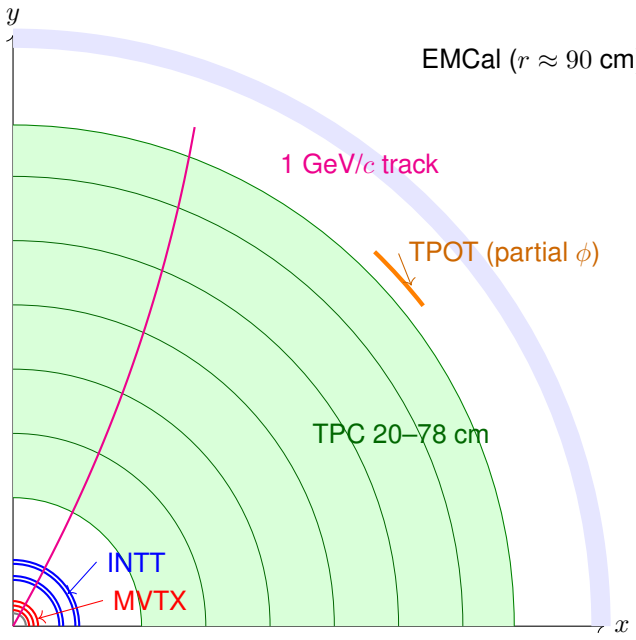


Figure 1: Schematic quarter view in the transverse (r–ϕ) plane, radii to scale (silicon layers drawn at approximate radii). The magenta curve is a toy $p_T = 1$ GeV/c track in 1.4 T (radius of curvature 2.4 m). TPOT is drawn schematically. Not an event display.

The combination is a "timing sandwich": the silicon detectors tell the TPC which crossing a track belongs to and where it started; the TPC supplies momentum; the silicon and TPOT together constrain the TPC distortions.

1.4 Layer numbering in software

In the sPHENIX tracking geometry, the layers are numbered from the inside out: MVTX layers 0–2, INTT layers 3–6, TPC layers 7–54 and TPOT layers 55–56 (check the current coresoftware geometry for any change). You will see these numbers in cluster keys and in residual plots.

Connect to your analysis

Your π^0/η analysis uses the MBD vertex. The tracking system can provide a vertex too, with much better z resolution, but only in events with enough tracks, and only for the crossing that the silicon identifies. Two practical links: (1) your photon candidates can be vetoed against tracks (a cluster matched to a track is a charged particle, not a photon); (2) the material of the trackers converts photons into e^+e^- pairs before they reach the EMCal (Section 10).

2 Charged particles in matter

Goal. Be able to estimate how much charge a minimum-ionising particle leaves in silicon and in gas, how much it scatters, and how it bends.

2.1 Energy loss: the Bethe–Bloch formula

A charged particle passing through matter loses energy mainly by kicking atomic electrons. A short classical argument gives the scaling [S10]. An electron at impact parameter b receives a momentum

kick $\Delta p = \int F\,\mathrm{d}t = 2ze^2/(bv)$ (Gauss's law for a straight-line passage at speed v). Its energy gain is $\Delta E = \Delta p^2/2m_e \propto z^2/(b^2v^2)$. Integrating over electrons in a shell $2\pi b\,\mathrm{d}b\,\mathrm{d}x$ between $b_{\min}$ and $b_{\max}$ gives a logarithm of $b_{\max}/b_{\min}$. The full relativistic quantum result is the Bethe–Bloch formula [R7]:

$$-\left\langle \frac{\mathrm{d}E}{\mathrm{d}x} \right\rangle = K z^2 \frac{Z}{A}\frac{1}{\beta^2}\left[\frac{1}{2}\ln\frac{2m_e c^2\beta^2\gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2}\right], \qquad K \approx 0.307\ \mathrm{MeV\,cm^2/mol}. \tag{1}$$

Its shape is what matters:

- at low momentum the $1/\beta^2$ term dominates, so slow particles lose much more energy;
- there is a broad minimum near $\beta\gamma \approx 3$–4: a particle there is a *minimum-ionising particle* (MIP);
- above it the loss rises slowly (relativistic rise), tamed by the density effect δ.

At fixed momentum, heavier particles have smaller β, so protons and kaons at $p < 1$ GeV/c ionise much more than pions. This is the basis of $\mathrm{d}E/\mathrm{d}x$ particle identification in the TPC.

2.2 Fluctuations: the Landau distribution

The energy lost in a thin layer fluctuates strongly, because rare hard collisions (producing energetic "delta electrons") carry a lot of energy. The distribution is asymmetric, with a long tail to high energy (Landau–Vavilov distribution). The most probable value (MPV) is noticeably below the mean. For a thin layer it is better to quote the MPV, and for $\mathrm{d}E/\mathrm{d}x$ in a TPC one uses a *truncated mean* (drop the highest samples) to suppress the tail.

Worked example: charge from a MIP in silicon

For silicon, $\langle \mathrm{d}E/\mathrm{d}x\rangle_{\min} \approx 1.66$ MeV cm^2/g $\times\, 2.33$ g/cm$^3 \approx 3.9$ MeV/cm $= 390$ eV/μm. Creating one electron–hole pair costs 3.6 eV. So the mean is about $390/3.6 \approx 108$ pairs per μm; the most probable value is lower, roughly 70–80 pairs per μm for thin sensors [S10, R7].

INTT (320 μm): mean $\approx 3.4\times10^4$ electrons, MPV $\approx 2.4\times10^4$. Easy to see above electronic noise of order 10^3 electrons.

MVTX: the charge is collected mostly from a thin (tens of μm) high-resistivity epitaxial layer, so a MIP gives of order 10^3 electrons, shared over a few pixels. The pixel threshold is of order 100 electrons [S10], well below this.

Worked example: electrons per cm in TPC gas

Typical total ionisation for a MIP (PDG table): neon ≈ 39 electrons/cm, $CF_4 \approx 100$/cm, argon ≈ 94/cm, isobutane ≈ 195/cm [R7]. A Ne:CF_4 50:50 mixture gives about 70 electrons/cm; an Ar:CF_4:iC_4H_{10} 75:20:5 mixture about 100/cm. A pad row about 1 cm long collects therefore of order 100 electrons in many small clusters. This is too little to read directly, so the TPC needs gas amplification of order 10^3 (Section 6).

2.3 Multiple Coulomb scattering

A charged particle also scatters elastically off nuclei. The net deflection after a layer of thickness x is roughly Gaussian with width [R7]

$$\theta_0 = \frac{13.6\ \mathrm{MeV}}{\beta c p}\, z\sqrt{\frac{x}{X_0}}\left[1 + 0.038\ln\frac{xz^2}{X_0\beta^2}\right]. \tag{2}$$

It scales as $1/p$ and as $\sqrt{x/X_0}$. At low momentum it dominates the track pointing resolution; this is why vertex detectors must be thin and close to the beam.

Worked example: scattering in the beam pipe and first MVTX layer

Take $x/X_0 \approx 0.5\%$ for the beam pipe plus one MVTX layer (each MVTX layer is about 0.3% X_0 [S10]). For $p = 1$ GeV/c: $\theta_0 = 13.6 \times 10^{-3} \times \sqrt{0.005} \times [1 + 0.038 \ln 0.005] \approx 0.77$ mrad. Projected over the distance to the first layer, $r_1 \approx 2.5$ cm, this gives about $2.5\ \text{cm} \times 0.77\ \text{mrad} \approx 19\ \mu$m. At 0.5 GeV/$c$ it is about 38 μm; at 5 GeV/c about 4 μm. Compare the MVTX hit resolution of a few μm [S10]: below a few GeV/c scattering, not pixel size, limits the DCA resolution. (Estimate.)

2.4 Motion in a magnetic field

In a solenoidal field B along z, a particle moves on a helix. Its transverse radius of curvature is

$$R\,[\text{m}] = \frac{p_T\,[\text{GeV}/c]}{0.3\,B\,[\text{T}]}. \tag{3}$$

In 1.4 T a 1 GeV/c track has $R = 2.38$ m; a 0.3 GeV/c track has $R = 0.71$ m. A track from the origin reaches radius r at an azimuthal deviation $\Delta\phi = \arcsin(r/2R)$ from its initial direction. Very soft tracks ($p_T \lesssim 0.15$ GeV/c, $2R < 0.78$ m) curl up inside the TPC ("loopers") [S10]. Positive and negative particles bend in opposite directions.

Worked example: a 1 GeV/c pion through the tracker (checkpoint exercise)

With $R = 2.38$ m, $\Delta\phi(r) = \arcsin(r/4.76\ \text{m})$:

Layer	MVTX 2.5	MVTX 3.9	INTT 7.2	INTT 10.3	TPC 20	TPC 78 cm
$\Delta\phi$ (deg)	0.30	0.47	0.87	1.24	2.41	9.43

The silicon hits lie almost on a straight line; the curvature is measured over the TPC. The sagitta over the TPC chord (about 0.58 m) is $s \approx L^2/8R = 1.8$ cm. A 5 GeV/c track has a sagitta of only 3.5 mm, which is why momentum resolution degrades with p_T. Expect 3 MVTX hits, 2 INTT hits (one in each barrel, from one of the two staggered sub-layers), up to 48 TPC clusters, and a TPOT hit only if the track points into a TPOT module.

3 Silicon detectors: from the pn junction to MAPS

Goal. Understand how a silicon sensor turns ionisation into a signal, and why strips, hybrid pixels and MAPS make different trade-offs.

3.1 The reverse-biased junction

Pure silicon has four valence electrons. Doping with a five-valence element (e.g. phosphorus) gives n-type material with free electrons (donors); doping with a three-valence element (e.g. boron) gives p-type material with free holes (acceptors) [S10]. At a pn junction, carriers diffuse across and recombine, leaving a *depletion zone* free of mobile charge, with a built-in electric field. A reverse bias (positive on the n side) widens the depletion zone. In a fully depleted sensor, an ionising particle creates electron–hole pairs which drift in the field to the electrodes; their motion induces the signal. The depletion width for a bias V is approximately [S10]

$$W(V) \approx \sqrt{\frac{2\varepsilon_0\varepsilon_{\text{Si}}V}{eN_D}} = \sqrt{2\varepsilon_0\varepsilon_{\text{Si}}\,\mu\,\rho\,V}, \tag{4}$$

with N_D the dopant concentration, ρ the resistivity and μ the carrier mobility.

Worked example: depletion widths
With $\rho = 20$ kΩ cm and $\mu = 1350$ cm^2/(V s): $W(1\ \mathrm{V}) \approx 75\ \mu$m and $W(45\ \mathrm{V}) \approx 500\ \mu$m. Because $W \propto \sqrt{V}$, doubling the depleted thickness needs four times the voltage. Thick strip sensors need tens to hundreds of volts; MAPS with a thin sensitive layer need only a few volts [S10].

3.2 Three silicon technologies

- **Strips.** Long, narrow electrodes (pitch 20–100 μm, length of cm). One coordinate is measured precisely, the other only to the strip length. Mature, cheap, reliable to manufacture [S10].
- **Hybrid pixels.** A pixelated sensor bump-bonded pixel by pixel to a readout chip. Two precise coordinates, fast, but bump bonding is delicate and adds material [S10].
- **Monolithic active pixel sensors (MAPS).** Sensor and readout electronics in the same CMOS chip; no bump bonds. Pixels can be small (20–30 μm), the chip can be thinned to tens of μm, and power is low. The price is speed (microsecond integration) [S10].

	Cell size	Speed	Thickness	Bias, power
Strips	20–100 μm $\times$ cm	fast	few hundred μm	high (100–500 V), high power
Hybrid pixels	50–100 μm	fast	few hundred μm	high, high power
MAPS	20–100 μm	slow (μs)	tens of μm	low (few V), low power

(Qualitative comparison following [S10].)

3.3 Position resolution of a binary cell

If a detector only says “hit” or “no hit” for a cell of pitch p, and a particle hits anywhere in the cell with flat probability, the rms error is [S10]

$$\sigma^2 = \frac{1}{p}\int_{-p/2}^{p/2} x^2\,\mathrm{d}x = \frac{p^2}{12}, \qquad \sigma = \frac{p}{\sqrt{12}}. \tag{5}$$

For the MVTX pixel ($\approx 29\ \mu$m) this gives about 8 μm; for the INTT strip pitch (78 μm) about 23 μm; along the INTT strip length (16 or 20 mm) about 4.6–5.8 mm. When charge is shared between neighbouring cells, the centroid of the cluster is more precise than $p/\sqrt{12}$; this is why multi-pixel clusters improve the resolution [S10]. The MVTX reaches a hit resolution of a few μm this way [S10].

3.4 Clustering

A particle usually fires more than one cell: it may cross a cell boundary, enter at an angle, or the charge diffuses. Clustering groups adjacent fired cells. It is naturally formulated as finding connected components of a graph whose nodes are fired cells and whose edges join neighbours [S10]. Hot (always-on) cells create fake clusters or bias real ones; dead cells split clusters; both must be masked [S10]. Clusters are larger for tracks at large $|\eta|$, which cross the sensor at a shallow angle and so traverse more silicon [S10].

3.5 Radiation damage

Radiation damages silicon by displacing lattice atoms (vacancies and interstitials, described by the non-ionising energy loss) and by ionisation in the oxide layers [S10]. The consequences are higher leakage

current, lower charge-collection efficiency, and (for thick sensors) a change of the depletion voltage. For MAPS, damage and environment shift the effective threshold and noise of the pixels, which are therefore monitored over time and re-tuned [S10].

4 The MVTX

Goal. Know the MVTX layout and ALPIDE readout well enough to explain its integration time, its noise masking, and its contribution to the DCA.

4.1 Layout and the ALPIDE chip

The MVTX has three layers of MAPS staves, at radii from about 2.5 cm to 4 cm, built on the ALICE ITS2 inner-barrel design [S10]. It uses the ALPIDE chip (ALice PIxel DEtector) [S10]:

Technology	180 nm CMOS MAPS
Sensor thickness	50 μm
Pixel size	$\approx 29 \times 27$ μm^2
Matrix	1024×512 pixels per chip
Material per layer (stave)	$\approx 0.3\%\, X_0$
Hit resolution	a few μm
Integration (strobe) time	several μs
Total channels	≈ 226 million pixels

(Design specifications as presented in [S10].) The front-end readout uses the ALICE Readout Unit, and the back end uses FELIX boards developed for ATLAS [S10]. The innermost layer sits only about 2.5 cm from the beam line, just outside the beryllium section of the beam pipe [S10].

4.2 How a pixel records a hit

Each pixel has an in-pixel amplifier and discriminator. When the collected charge exceeds the threshold (of order 100 electrons [S10]), the discriminator output goes high for a time that depends on the charge (microseconds). The chip latches a hit only if this output overlaps a **strobe** window [S10]. There are three ways to operate:

- **Streaming mode:** strobes are issued continuously, back to back (windows of order 10 μs with a short gap). Every hit overlaps some strobe and is recorded [S10].
- **Triggered mode:** a short strobe is issued only after a trigger. Because the trigger arrives a few μs after the collision (latency), the hit is recorded only if the discriminator is still high when the strobe arrives [S10].
- **Extended (triggered) mode:** a long strobe is opened on a trigger, giving the robustness of streaming within a window [S10].

Pitfalls

In triggered mode, hits whose discriminator pulse is shorter than the trigger latency are lost. The time above threshold depends on the deposited charge, so the loss is not uniform: it can depend on particle type, momentum and η (large-$|\eta|$ tracks cross more silicon; slow, heavily ionising particles deposit much more charge) [S10]. An efficiency measured in streaming mode does not automatically apply to triggered data.

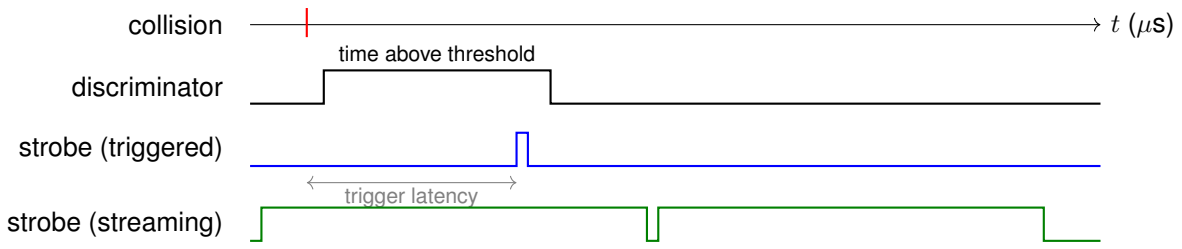


Figure 2: Schematic ALPIDE timing (toy durations). A hit is latched if the discriminator output overlaps a strobe. In triggered mode a short strobe arrives after the trigger latency; if the discriminator has already returned low, the hit is lost. In streaming mode back-to-back strobes always overlap.

4.3 Data reduction

The MVTX is binary: each pixel is on or off; no amplitude is stored [S10]. Only fired pixels are sent (zero suppression). In a typical event only a tiny fraction of the pixels fire, of order 10^{-4} [S10]. The address encoding is optimised: with 512 rows and 1024 columns, a pixel address needs $9 + 10 = 19$ bits. Eight hits would need 152 bits if sent one by one, but if one sends the first address and then a small bitmap of which neighbours are also hit, a cluster of 8 pixels costs only a few tens of bits [S10].

4.4 Noisy pixels and masking

Because all signals look the same (a fired pixel), a few stuck or noisy pixels per chip can produce as many hits as real particles [S10]. In a chip with half a million pixels and a few tens of real hits per event, a couple of dozen permanently-on pixels would hide the signal. Noisy pixels are therefore identified in dedicated runs and masked [S10]. Their number changes with time (radiation, settings), so masks are part of the conditions.

4.5 Impact-parameter resolution

A useful approximation for the DCA resolution of a two-layer telescope is

$$\sigma_{\mathrm{DCA}}^2 \approx \sigma_{\mathrm{geom}}^2 + \left(\frac{r_1 \cdot 13.6\ \mathrm{MeV}}{p_T\, c} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \frac{x}{X_0} \right] \right)^2 , \tag{6}$$

where σ_{geom} depends on the hit resolutions and the lever arm between layers, r_1 is the radius of the first measurement and x/X_0 the material before it [S10]. Both terms favour small r_1 and little material: hence the MVTX at 2.5 cm with 0.3% X_0 per layer. From the earlier worked example the scattering term is about 20 μm at 1 GeV/c, so the MVTX DCA resolution at low p_T is set by scattering. This DCA precision is what allows heavy-flavour hadrons (with $c\tau$ of order 100–500 μm) to be separated from prompt particles.

4.6 Vertexing with the MVTX

Once tracks exist, the primary vertex is found by extrapolating them back and finding where they meet [S10]. The beam spot is small and stable in x–y (order 100 μm) but long in z (tens of cm) [S10]. A standard procedure: fit a vertex; remove tracks that are obviously incompatible (large DCA); test the effect of removing each track in turn on the vertex χ^2, and drop tracks that change it most. This rejects tracks from displaced decays [S10].

4.7 Alignment

Mechanical survey (coordinate-measuring machines, laser scans) gives positions to some tens of μm, but the detector moves when the magnet is on or when it warms up by 10–20 K during operation [S10]. Cosmic rays (straight tracks, especially with the field off) and dedicated early-run data are used to align the sensors in situ [S10]. Alignment procedures are treated in Module 8.

Checks

- Chip hit maps: after masking, the occupancy should be smooth; isolated very high bins are noisy pixels.
- Cluster size vs. η: should increase with $|\eta|$ [S10].
- Threshold and noise per layer versus time: drifts indicate radiation effects or settings changes [S10].
- Strobe/BCO bookkeeping: in streaming data, check which strobe a hit belongs to before associating it with a trigger.

5 The INTT

Goal. Know the INTT layout and readout, and understand why its single-crossing timing is the key to using the slow MVTX and TPC data.

5.1 Layout

The INTT is a two-layer barrel of silicon strip ladders covering the full azimuth and $|\eta| < 1.1$, between the MVTX and the TPC [S11, R1, R6].

- 56 ladders: 24 in the inner barrel, 32 in the outer barrel [S11].
- Each barrel consists of two staggered sub-layers, so that the gaps of one are covered by the other; the four sub-layer radii are about 7.2, 7.8, 9.7 and 10.3 cm [S11].
- Sensors are 320 μm thick with 78 μm strip pitch [S11, R1].
- Two sensor types along z: type A with 16 mm strips and type B with 20 mm strips [S11, R1].
- Strips run along z, so the INTT measures $r\phi$ precisely (≈ 23 μm binary) and z only to the strip length.

Small longitudinal gaps come from the inactive edges of the sensors, and small azimuthal gaps from the two detector halves not closing perfectly [S11].

5.2 Channels and readout

Each half-ladder carries one type-A and one type-B sensor and is read by FPHX chips, one per block of 128 strips; the FPHX chip and much of the design come from the PHENIX FVTX [S11, R1]. Counting [S11]:

- type A: $128 \times 16 = 2048$ channels; type B: $128 \times 10 = 1280$; so 3328 channels per half-ladder;
- 112 half-ladders, so $3328 \times 112 = 372\,736$ channels in total.

The signal path is: sensor → FPHX on the high-density interconnect (HDI) → bus extender and conversion cable → readout card (ROC, re-used from PHENIX FVTX) → FELIX servers in the counting house [S11, R1]. There are 16 ROCs (8 north, 8 south) and 8 FELIX servers [S11]. The FPHX provides a hit with a coarse amplitude (a few-bit ADC) and its beam-crossing number [R1].

5.3 Timing: one crossing

The INTT readout records each hit with the granularity of one RHIC crossing (106 ns) [R3]. Its intrinsic time resolution is much better than the crossing spacing, so after the timing is tuned in commissioning (a delay scan in steps of a fraction of a crossing), essentially all hits from a collision fall into one crossing [S11, R3]. This makes the INTT the detector that assigns a track to its bunch crossing [R2, R3].

5.4 Triggered and streaming operation

The FELIX-based readout can run triggered or streaming [S11]:

- **Triggered (extended)** readout, used in Au+Au and part of p+p: on a trigger, hits are read for a window of about 100 crossings (about 10 μs) [S11].
- **Streaming** readout, used in p+p and O+O: hits are recorded continuously; the crossing counter of the FPHX is reset every RHIC revolution at the abort gap, so a "strobe" covers one turn of the beam [S11].

5.5 Performance checks with physics

- **Good-channel selection:** hot, cold and dead channels are identified in data and removed; in the public Au+Au multiplicity analysis they were about 4% of the channels [R6].
- **Synchronisation:** the numbers of clusters in the inner and outer layers must be correlated event by event; so must the INTT cluster count and the MBD charge sum. A broken correlation reveals desynchronised servers or subsystems [S11].
- **Vertex:** the INTT z vertex (from tracklets) correlates with the MBD z vertex; the transverse beam position varies with z because of the beam crossing angle [S11].
- **Hit efficiency:** the public beam test found a signal-to-noise ratio above 15 and a hit efficiency above 99% [R2]; the installed detector reproduces this [R3].

5.6 How a hit efficiency is measured (method)

The flash talk [S28] illustrates a general method that you can reuse for any layer:

1. Choose a *target* layer (here the inner INTT). Reconstruct tracks *without* it (MVTX triplet + outer INTT).
2. Select clean tracks: a good primary vertex (several tracks), small DCA, isolation from other tracks, unshared clusters.
3. Project each track to the target layer. Keep only projections inside the fiducial area (away from chip edges, dead or hot channels).
4. Search for a cluster within a residual window (only in ϕ, because the z resolution is coarse) and in the right time bucket.
5. Efficiency = (tracks with a matched cluster) / (tracks projected into the fiducial area), per half-ladder.

The method measures the *hit* efficiency of the active area, not tracking efficiency or acceptance. Two subtleties: an impure track sample biases the efficiency low; and hits can leak into the neighbouring crossing or be "carried over" from the previous turn, so the time-bucket choice matters [S28].

Connect to your analysis

The same tag-and-probe logic appears in your analysis: the trigger efficiency of the Photon trigger is measured on events taken with an independent (minimum-bias) trigger, and your single-photon efficiency is defined relative to a clean reference. In each case: define the probe sample independently of the thing you measure; restrict to a fiducial region; and check purity.

Pitfalls

- The INTT z coordinate is only known to the strip length (16 or 20 mm). Do not expect mm-level z from INTT-only tracklets; the vertex z resolution comes from combining many tracklets.
- Do not compare INTT occupancies across runs without the good-channel map for each run.
- In streaming data, hits from neighbouring crossings are present; a cut on the crossing number relative to the trigger is required.

6 Gas detectors: drift, diffusion and amplification

Goal. Understand the physics that sets the TPC's precision and its main difficulty, space charge.

6.1 From wire chambers to micro-pattern detectors

In a gas detector, a charged particle ionises gas atoms; the freed electrons drift in an electric field to a region of very high field where they multiply in an avalanche, producing a measurable signal [S12].

- In a **Geiger–Müller** tube the avalanche runs away (photons from the avalanche ionise elsewhere): a large signal, but independent of the deposited energy and with long dead time [S12].
- Adding a **quencher** (a polyatomic gas that absorbs energy in vibrations and rotations) breaks the feedback loop and gives a **proportional** signal [S12].
- Multi-wire proportional chambers and drift chambers measure positions with arrays of wires; the **TPC** drifts the charge over a long distance to a segmented readout plane and measures the third coordinate from the drift time [S12].
- Modern amplification stages are micro-pattern gas detectors (MPGDs): **GEMs** (thin perforated foils with a high field in the holes) and **Micromegas** (a micro-mesh above the pads with a very narrow, high-field gap) [S12].

6.2 Drift

In a field $\vec{E}$, electrons reach a constant drift velocity v_d, set by collisions with gas molecules. With a magnetic field $\vec{B}$, the Langevin equation gives

$$\vec{v}_d = \frac{\mu E}{1+\omega^2\tau^2}\left[\hat{E} + \omega\tau\,\hat{E}\times\hat{B} + \omega^2\tau^2(\hat{E}\cdot\hat{B})\hat{B}\right], \qquad \omega = \frac{eB}{m_e}, \tag{7}$$

where τ is the mean time between collisions and $\mu = e\tau/m_e$ the mobility. If $\vec{E} \parallel \vec{B}$ (the ideal TPC), the drift is straight along the field lines. Any component of $\vec{E}$ perpendicular to $\vec{B}$ (from field-cage imperfections, the

field of accumulated ions, or non-uniformity of $\vec{B}$) makes the electrons drift sideways ("$E \times B$ distortion"). The sPHENIX gas has a high drift velocity, about 8 cm/μs in the Ne:CF_4 design mixture [R5].

6.3 Diffusion

A drifting electron cloud spreads by diffusion: $\sigma = D\sqrt{L}$ for drift length L, with D the diffusion coefficient (in $\mu\text{m}/\sqrt{\text{cm}}$). A magnetic field parallel to $\vec{E}$ reduces the *transverse* diffusion by a factor of about $1/\sqrt{1+\omega^2\tau^2}$, because electrons spiral around the field lines. CF_4 mixtures combine a high drift velocity with a large $\omega\tau$, hence low transverse diffusion in 1.4 T [R4, R5]. Low diffusion is good for resolution, but if the cloud becomes narrower than a pad, most hits land on a single pad and the position information is lost; this is why sPHENIX uses **zigzag pads** that share charge between neighbours (Section 7) [S12].

6.4 Gas amplification

In a high field, an electron gains enough energy between collisions to ionise; the number of electrons grows as $\mathrm{d}N = \alpha N \,\mathrm{d}x$ (Townsend coefficient α), so the gain is $G = \exp\int \alpha \,\mathrm{d}x$. A stack of several GEM foils shares the gain among stages, which makes operation stable (lower sparking risk) and allows the ion back-flow to be minimised [S12].

6.5 Ion back-flow and space charge

Each avalanche produces as many positive ions as electrons. Ions are about a thousand times slower than electrons. If they flow back into the drift volume (**ion back-flow**, IBF), they accumulate and distort the drift field [S12]. The ion density in the drift volume comes from two sources: primary ionisation from tracks, and back-flowing ions from the amplification.

Worked example: why IBF must be below about 1%

Let n_p be the primary electrons per unit time. The back-flowing ions per unit time are $n_p \times G \times \epsilon_{\text{IBF}}$. With $G = 2000$ and $\epsilon_{\text{IBF}} = 1\%$, each primary electron sends back 20 ions: the back-flow is 20 times the primary ionisation. With $\epsilon_{\text{IBF}} = 0.2\%$ it is 4 times. Ions drift slowly: with a toy mobility of 2–4 $\text{cm}^2/(\text{V s})$ in a 400 V/cm field they move 8–16 m/s, so they need of order 0.1 s to cross 1 m. At a collision rate of tens of kHz, the ions of thousands of events are in the volume at any time. The distortion is therefore set by the average luminosity and the IBF, and it fluctuates with the beam. (Toy numbers.)

The sPHENIX TPC uses a quadruple-GEM stack operated in a low-IBF configuration (following the ALICE TPC upgrade), with an IBF of order 1% or below [S12, R4, R5]. Because the drift field is ungated (no gating grid closes after each event), the TPC can read continuously [R5].

6.6 Gas choice

The design gas was Ne:CF_4, chosen for high drift velocity, low transverse diffusion and fast ion drift [R4, R5]. Different Ne:CF_4 ratios trade diffusion against electron attachment (loss of drifting electrons to CF_4 in high-field regions) [S12]. When the war in Ukraine disrupted the world neon supply and its price rose sharply, the collaboration re-optimised the gas and settled on an argon-based mixture, Ar:CF_4:isobutane (75:20:5), which mimics the Ne:CF_4 properties and proved robust against discharges from highly ionising particles [S12].

7 The sPHENIX TPC

Goal. Know the TPC geometry, readout and calibration tools, and how its design sets momentum resolution and pile-up.

7.1 Design philosophy

The TPC is the main tracker and provides the momentum resolution [S12]. Three design principles [S12]:

1. **Large lever arm:** maximise the active radial range, so make the field cages as thin as possible.
2. **High precision:** good single-point resolution from a low-diffusion gas and charge-sharing zigzag pads.
3. **High accuracy:** low distortions, from low IBF and from calibration tools (lasers, and silicon and TPOT as external references).

7.2 Geometry and mechanics

- Active radius about 20 to 78 cm, $|\eta| < 1.1$; the outer radius is smaller than the inner radius of the ALICE TPC [R4, R5].
- A central membrane at $z = 0$ at high voltage divides the TPC into two halves; electrons drift about 1.05 m to each endcap in a field of about 400 V/cm [R4, R5].
- The maximum drift time is $105\ \mathrm{cm}/(8\ \mathrm{cm}/\mu\mathrm{s}) \approx 13\ \mu$s.
- Each endcap ("wagon wheel", one piece of aluminium) holds 12 sectors, each with three radial modules R1, R2, R3, i.e. 72 modules in total; each module has 16 pad rows, so there are 48 pad rows (layers) [S12].
- The field cages are honeycomb/carbon-fibre composites with Kapton insulation and printed field-shaping stripes connected by resistor chains; the stripes make the drift field uniform close to the cage walls [S12].
- The mechanics were designed to survive a magnet quench (eddy currents, deflections) [S12].

7.3 Readout: GEMs, zigzag pads, SAMPA, FELIX

- **GEM stack.** Four GEM foils per module amplify the drift electrons. The foils are segmented to limit the energy of a discharge [S12].
- **Zigzag pads.** The pads under the GEMs interleave like the teeth of two combs; each pad's tips reach deep into the neighbour ("incursion"). With a narrow charge cloud, rectangular pads would often see the charge on one pad only (position known only to $p/\sqrt{12}$); zigzags share the charge and give a resolution that is a small fraction of the pitch. Laboratory studies showed that incursions close to 100% and very small gaps between pads give the best linearity [S12, R4].
- **Front-end.** The signals are shaped and digitised continuously by the SAMPA chip (developed by ALICE), in a version modified for sPHENIX with a faster peaking time [S12, R4, R5]. Each front-end card (FEE) reads 256 pads; a sector is served by roughly two dozen FEEs, i.e. of order 600 FEEs for the whole TPC (the exact count is in the TPC documentation) [S12].
- **Back-end.** Data aggregation modules (DAMs) based on FELIX boards, one per sector (24 in total), each in an event-buffer and data-compressor (EBDC) server [S12].

7.4 What the TPC measures

For each cluster the TPC provides r (the pad row), ϕ (the charge centroid across pads) and z (from the drift time). Note that z is measured as $z = z_{\rm end} \mp v_d\,(t - t_0)$: it requires the collision time t_0. The drift velocity v_d must be known everywhere, and t_0 must come from another detector.

Worked example: wrong crossing means wrong z

If a track is assigned to a crossing one bunch too early or too late, t_0 is off by 106 ns and all its TPC clusters move by $v_d \times 106\ \mathrm{ns} \approx 8\ \mathrm{cm}/\mu\mathrm{s} \times 0.106\ \mu\mathrm{s} \approx 8.5$ mm in z — in opposite directions on the two sides of the central membrane. A track crossing the membrane then appears broken, and its TPC segment no longer points to the silicon hits. This is why the INTT crossing assignment is essential [S10, S11].

7.5 Momentum resolution

For N equally spaced points with resolution σ_x over a lever arm L in a field B, the curvature measurement gives (Gluckstern) [R7]

$$\left(\frac{\sigma_{p_T}}{p_T}\right)_{\rm meas} = \frac{\sigma_x\, p_T}{0.3\, B\, L^2}\sqrt{\frac{720}{N+4}}, \tag{8}$$

and multiple scattering adds a constant term $(\sigma_{p_T}/p_T)_{\rm ms} \approx \frac{0.0136}{0.3BL\beta}\sqrt{x/X_0} \times \mathcal{O}(1)$.

Worked example: TPC momentum resolution (estimate)

$B = 1.4$ T, $L = 0.58$ m, $N = 48$. For a toy $\sigma_x = 150$ μm: $(\sigma_{p_T}/p_T)_{\rm meas} = 1.5 \times 10^{-4}\, p_T/(0.3 \times 1.4 \times 0.336) \times \sqrt{720/52} \approx 0.4\% \times p_T$. So about 0.4% at 1 GeV/c and 4% at 10 GeV/c from the TPC alone; for $\sigma_x = 250$ μm these become 0.7% and 7%. The scattering term, with about 1% X_0 of material over the lever arm, is about 1%. Adding the silicon points increases the lever arm and improves high-p_T resolution. This is the reasoning behind the Υ mass-resolution requirement that drove the TPC design [R4]. (Estimate; not a performance number.)

7.6 Distortions and calibration tools

The ideal TPC has uniform $\vec{E} \parallel \vec{B}$ and uniform v_d. Reality has [S12]:

- a non-uniform magnetic field, especially near the ends of the solenoid;
- space charge from primary and back-flowing ions, which varies with luminosity and in time;
- imperfections of the field cage and the central membrane, and misalignment of the field cage relative to the endcaps and to $\vec{B}$.

Distortions can reach the millimetre-to-centimetre scale, far larger than the intrinsic resolution, so they must be measured and corrected [S12]. The tools are:

1. **Diffuse laser on the central membrane.** The membrane carries a pattern of small aluminium dots. UV laser light releases photo-electrons from them at known positions; these drift the full length to the readout. The measured pattern shows the integrated distortion and the absolute integrated drift velocity [S12].
2. **Line lasers.** UV laser beams steered by mirrors into the volume ionise the gas along straight lines that mimic straight tracks; comparing measured to true straight lines maps distortions throughout the volume. Comparing beam-on and beam-off isolates space charge; field-on and field-off isolates $E \times B$ effects [S12].

3. **External references.** The MVTX/INTT inside and TPOT outside are not affected by drift distortions; the residuals between TPC track segments and these detectors measure distortions in data [S12].

Signs that distortions are under control include: residuals in z and $r\phi$ that do not grow with luminosity; consistent K^0_S and Λ masses from all regions and collision systems; and equal momentum scales for positive and negative particles (space-charge distortions shift opposite charges in opposite directions) [S12]. The calibration procedures are the subject of Module 8.

Pitfalls

- TPC z positions depend on t_0: never use TPC-only tracks to determine which crossing a track belongs to.
- A momentum-scale difference between $+$ and $-$ tracks is a classic symptom of residual distortions.
- Distortion corrections depend on the collision rate: a correction derived in a low-luminosity period does not apply to a high-luminosity period.

8 TPOT: the TPC Outer Tracker

Goal. Understand why a small Micromegas detector is the external reference for TPC distortions.

A Micromegas detector has a conversion/drift gap of a few mm above a micro-mesh and a very narrow amplification gap (about a hundred μm) between mesh and readout strips [S12]. Electrons drift only a few mm before amplification, so drift distortions are negligible; the measured position is the true position [S12]. TPOT sits just outside the TPC. Because of funding, it covers only limited azimuthal regions; given that constraint, the chosen layout covers the full z length (four chambers along z) over as much azimuth as possible [S12]. Each TPOT module provides two coordinates (strips measuring ϕ and strips measuring z). Tracks passing through the silicon, the TPC and TPOT measure the TPC distortion at the outer radius, from where it can be extrapolated with the help of the laser data. In Hemmick's view, extending TPOT to full azimuth would have been the single most useful improvement to sPHENIX [S12].

9 Integration times, pile-up and the timing sandwich

Goal. Be able to say, for each detector, which collisions it sees together with the triggered one.

Detector	Time window	Consequence
INTT	one crossing (106 ns)	identifies the crossing of each hit/track
MVTX	strobe of several μs (≈ 10 μs in streaming)	sees all collisions in the strobe; needs INTT to separate them
TPC	drift up to ≈ 13 μs	sees collisions from ± 13 μs around the trigger; z needs t_0
TPOT	short drift, continuous	distortion-free reference
EMCal/HCal	a few samples around one crossing (Module 3)	triggered readout, single crossing

Worked example: TPC pile-up (checkpoint exercise)

The TPC records charge from all collisions that happened within one drift time before or after the triggered one. The number of extra collisions is $N \approx R \times 2 \times 13\ \mu\mathrm{s}$ for an interaction rate R. Toy rates: $R = 50$ kHz gives $N \approx 1.3$; $R = 500$ kHz gives $N \approx 13$. Each extra collision's tracks appear shifted in z by $v_d \Delta t$ and do not point to the silicon hits of the triggered crossing. Pattern recognition must therefore start from silicon seeds (MVTX+INTT) in the right crossing and extend them into the TPC, or match TPC segments to silicon segments. (Toy rates; see Module 2 for actual collision rates.)

The combination works as a sandwich [S10, S11, S12]:

1. INTT hits define the crossing.
2. MVTX hits in the corresponding strobe, combined with INTT hits, form silicon seeds with precise pointing.
3. TPC track segments are matched to the seeds; the crossing fixes t_0 and hence the TPC z.
4. The full track is fitted (ACTS Kalman filter, Module 8) with all points.

For triggered running the MVTX and INTT readout windows are extended to cover the trigger latency and the TPC drift; in streaming running all detectors except the calorimeters record continuously, and the association happens offline (Module 5) [S10, S11].

9.1 Raw data to clusters (software pointers)

In the offline software, each subsystem has a raw-data decoder that turns DAQ packets into hits in `TrkrHitSet` containers (node `TRKR_HITSET`), and a clusterizer that groups hits into `TrkrCluster` objects (node `TRKR_CLUSTER`). Examples of module names in coresoftware are `MvtxClusterizer`, `InttClusterizer`, `TpcClusterizer` and `MicromegasClusterizer`, with corresponding combined raw-data decoders for each subsystem (check the current `offline/packages` directories for exact names). Modules 6 and 8 show how these are chained in a macro.

10 Roles of each detector, and what they mean for photons

Goal. Summarise the roles and connect tracking-detector properties to calorimeter analyses.

- **Vertexing:** MVTX (precision), INTT (crossing and coarse z), TPC (pattern recognition). Global vertex combines tracks and the MBD (Module 2).
- **Momentum:** TPC, with silicon and TPOT extending the lever arm.
- **DCA and heavy flavour:** MVTX.
- **Timing and pile-up rejection:** INTT.
- **Particle identification:** TPC $\mathrm{d}E/\mathrm{d}x$ at low momentum.
- **Multiplicity:** INTT tracklets (used for the public Au+Au $\mathrm{d}N_{\mathrm{ch}}/\mathrm{d}\eta$ result [R6]).
- **Calibration of the TPC:** silicon and TPOT as distortion-free references.
- **Track–calorimeter matching:** E/p for electrons, charged-track veto for photons, particle-flow jets (Modules 7, 8).

10.1 Photon conversions in the tracker material

A photon converts into e^+e^- with probability $P = 1 - \exp(-\frac{7}{9}\, x/X_0)$. The public multiplicity paper gives about 3.8% X_0 between the collision point and the INTT (1.8% from the beam pipe and 2.0% from the MVTX), and the INTT itself adds about 3.7% X_0 according to [S11]; the TPC field cages and gas add more [R6, S11].

Worked example: conversion probability before the EMCal

For $x/X_0 = 3.8\%$, $P = 1 - e^{-0.778\times 0.038} \approx 2.9\%$. For a total of 8% X_0 up to the EMCal (toy value including the INTT and TPC), $P \approx 6\%$; for 10%, $P \approx 7.5\%$. For a $\pi^0 \to \gamma\gamma$ the probability that at least one photon converts is about twice this. A converted photon becomes an e^+e^- pair that the 1.4 T field opens up in ϕ; the EMCal may then see a wider cluster, two clusters, or a cluster with a track pointing at it. (Toy material totals.)

Connect to your analysis

- Conversions are a few-percent effect on your π^0/η efficiency and on the photon energy scale. Your PYTHIA8 + Geant4 efficiency includes them only if the material in the simulation is correct; a material uncertainty is therefore a systematic on the cross section.
- If you ever apply a charged-track veto to photon clusters, converted photons (whose $e^\pm$ are tracked) will be vetoed too; this must be in the efficiency.
- Your analysis uses the MBD vertex for photon kinematics. A vertex error δz changes each photon's polar angle by $\delta\theta \approx \delta z\, \sin^2\theta / R_{\rm EMCal}$ (about 0.1 rad for $\delta z = 10$ cm at $\theta = 90^\circ$, $R = 93$ cm). Both photons of a pair shift by nearly the same amount, so the opening angle and the mass change only at second order, but each photon's $E_T = E\sin\theta$ and the pair rapidity change at first order. A precise vertex therefore matters for the p_T assignment and the $|z_{\rm vtx}|$ acceptance.

11 Summary

- A MIP gives about 108 electron–hole pairs per μm (mean) in silicon and of order 100 electrons per cm in the TPC gas; gas needs amplification, silicon does not.
- MVTX: three ALPIDE MAPS layers at 2.5–4 cm, $\approx 29 \times 27$ μm^2 pixels, 0.3% X_0 per layer, few-μm hits, several-μs strobes; sets the DCA resolution, which is scattering-dominated at low p_T.
- INTT: 56 strip ladders in two barrels (four sub-layers) at 7–10 cm, 78 μm pitch, 16/20 mm strips, 372 736 channels; single-crossing timing ties tracks to their bunch crossing.
- TPC: 20–78 cm, 48 pad rows, quad-GEM low-IBF readout, zigzag pads, SAMPA continuous readout, ≈ 8 cm/μs drift and ≈ 13 μs maximum drift; provides momentum; suffers space-charge distortions, measured with lasers and with silicon/TPOT.
- TPOT: Micromegas just outside the TPC with negligible drift; external distortion reference.
- Timing sandwich: INTT crossing $\to$ silicon seeds $\to$ TPC matching (t_0 fixes z) $\to$ full fit.

12 Hands-on exercises on SDCC (optional, suggested)

These exercises are optional. They have not been run as part of writing this module; names and commands are suggestions to be checked against the current software. The text does not depend on them.

1. **Sketch a 1 GeV/c pion (checkpoint).** Using $R = p_T/(0.3B)$ and $\Delta\phi = \arcsin(r/2R)$, draw the transverse view of a π^+ and a π^- of $p_T = 1$ GeV/c through the MVTX, INTT and TPC radii, and mark where TPOT would be. Repeat for 0.3 and 5 GeV/c.
2. **TPC drift and pile-up (checkpoint).** Compute the maximum drift time for $v_d = 8$ cm/μs and drift length 105 cm. For three interaction rates of your choice, compute the mean number of out-of-time collisions in the TPC readout window, and the z shift of a track assigned to the wrong crossing by ± 1 and ± 5 crossings.
3. **Resolution calculator.** Code the Gluckstern and scattering formulas and plot σ_{p_T}/p_T vs. p_T for $\sigma_x = 100$–300 μm. Then code the DCA formula and find the p_T where the scattering term equals a 5 μm geometric term.
4. **Look at clusters in a DST.** Set up the environment with

   ```
   source /opt/sphenix/core/bin/sphenix_setup.sh -n new
   ```

   take one tracking DST of a Run-24 p+p run from the FileCatalog (Module 5), and in a small `SubsysReco` loop over the `TRKR_CLUSTER` container; fill the cluster radius and the layer number to see the MVTX/INTT/TPC/TPOT structure. Check the current container interface in coresoftware (`TrkrClusterContainer`, `TrkrDefs`) before writing the loop.
5. **Conversion probability.** Using $P = 1 - \exp(-7x/9X_0)$, compute the conversion probability before the EMCal as a function of the assumed material and the probability that a π^0 has at least one converted photon. Compare with how your PYTHIA8 efficiency changes if you vary the material by $\pm 10\%$ (conceptually; no simulation needed).

13 Self-test questions

1. Why do slow protons and kaons deposit more energy per unit length than pions of the same momentum?
2. Why is the most probable energy loss in a thin silicon sensor lower than the mean?
3. Estimate the number of electron–hole pairs a MIP creates in a 320 μm INTT sensor.
4. Derive the $p/\sqrt{12}$ resolution of a binary cell. Why can a real pixel detector do better?
5. List two advantages and one disadvantage of MAPS compared with hybrid pixels.
6. Explain how an ALPIDE hit is latched, and why triggered-mode operation can lose hits from heavily ionising particles.
7. Why does the DCA resolution at $p_T = 1$ GeV/c depend more on material than on pixel size?
8. What is the INTT's most important contribution to sPHENIX tracking, and why can the MVTX and TPC not provide it?
9. Give the INTT channel count from the sensor geometry.
10. What happens to the z positions of a TPC track if its crossing is misassigned by one bunch?
11. Why does the TPC need a low ion back-flow? Give a quantitative argument.
12. What are zigzag pads for?
13. Name the three tools used to measure TPC distortions and say what each measures.
14. Why is TPOT a good reference for TPC distortions, and what limits its usefulness?
15. Estimate the TPC momentum resolution at 5 GeV/c for $\sigma_x = 150$ μm, ignoring scattering.

A Answers

1. At equal momentum, heavier particles have smaller β. Bethe–Bloch rises as $1/\beta^2$ at low $\beta\gamma$, so below about 1 GeV/c kaons and protons are well above the MIP plateau while pions are near minimum.
2. The distribution has a long high-energy tail from rare, hard collisions producing delta electrons; these pull the mean up but rarely happen, so the most probable value lies below the mean.
3. Mean $\approx 108 \times 320 \approx 3.4 \times 10^4$ pairs; the most probable value $\approx 2.4 \times 10^4$ (using ≈ 75 per μm).
4. With flat probability across $[-p/2, p/2]$, $\langle x^2 \rangle = \frac{1}{p}\int x^2 \, \mathrm{d}x = p^2/12$. Real pixels share charge between neighbours (diffusion, angled tracks), and the cluster centroid locates the crossing within the cell.
5. Advantages: small pixels, very thin (low material), no bump bonds, low bias voltage and power. Disadvantage: slow (microsecond integration), so many crossings are integrated together; also, historically, cost of a new technology.
6. A hit is recorded if the in-pixel discriminator output (high while the signal is above threshold) overlaps a strobe window. In triggered mode the strobe arrives after the trigger latency of a few μs. Whether the discriminator is still high then depends on the deposited charge, so the efficiency depends on particle type, momentum and η.
7. The scattering term is $r_1\theta_0$, with $\theta_0 \propto \sqrt{x/X_0}/p$; at 1 GeV/$c$ it is about 20 μm, larger than the few-μm geometric term from pixel size and lever arm. Less material or smaller r_1 helps more than smaller pixels.
8. Single-crossing time stamps. The MVTX integrates over a strobe of several μs and the TPC over a drift time of ≈ 13 μs, so both see many crossings; only the INTT tells which crossing a track came from.
9. Per half-ladder: type A $128 \times 16 = 2048$, type B $128 \times 10 = 1280$, total 3328. With 112 half-ladders: $3328 \times 112 = 372\,736$.
10. All clusters shift by $v_d \times 106$ ns ≈ 8.5 mm in z, in opposite directions in the two TPC halves; the track no longer points at its silicon hits and breaks at the central membrane.
11. Back-flowing ions per primary electron equal $G\epsilon_{\mathrm{IBF}}$: 20 for $G = 2000$ and 1% IBF. Slow ions accumulate over many events and distort the drift field. Keeping ϵ_{IBF} at or below 1% keeps the space charge to a level that can be corrected.
12. In a low-diffusion gas the charge cloud is narrow; with rectangular pads it would often land on one pad, giving only $p/\sqrt{12}$ resolution. Interleaved zigzag pads share the charge among neighbours, so the centroid gives a resolution much smaller than the pitch.
13. Diffuse laser on the central-membrane dot pattern (integrated distortion over the full drift, absolute drift velocity); line lasers (straight reference lines throughout the volume; beam on/off and field on/off separate space charge and $E \times B$); external references MVTX/INTT and TPOT (residuals of TPC segments with respect to distortion-free detectors in real data).
14. Its drift gap is only a few mm, so its measured positions are free of drift distortions. Its usefulness is limited by its small azimuthal coverage: it measures distortions only in the sectors it covers.
15. $0.4\% \times 5 \approx 2\%$ (from $\sigma_x p_T/(0.3BL^2)\sqrt{720/(N+4)}$ with $B = 1.4$ T, $L = 0.58$ m, $N = 48$).

B Glossary

MIP Minimum-ionising particle, $\beta\gamma \approx 3$–4.

Bethe–Bloch Formula for the mean ionisation energy loss.

Landau distribution Asymmetric distribution of energy loss in thin layers.

Multiple scattering, θ_0 Random angular deflection from nuclear Coulomb scattering.

Depletion zone Region of a pn junction free of mobile carriers, where signal charge is collected.

MAPS Monolithic active pixel sensor; sensor and electronics in one CMOS chip.

ALPIDE The MAPS chip of ALICE ITS2, used in the MVTX.

Strobe Time window during which the ALPIDE latches hits.

DCA Distance of closest approach to the vertex (impact parameter).

BCO Beam-clock counter; the bunch-crossing number.

FPHX Readout chip of the INTT (from PHENIX FVTX).

FELIX Readout board/server used by MVTX, INTT, TPC and TPOT.

Drift velocity v_d Constant speed of electrons in the TPC gas.

Diffusion Spreading of the drifting charge cloud, $\sigma = D\sqrt{L}$.

GEM Gas electron multiplier foil.

Micromegas Micro-mesh gaseous detector with a narrow amplification gap.

IBF Ion back-flow, the fraction of avalanche ions entering the drift volume.

Space charge Accumulated positive ions that distort the drift field.

Zigzag pads Interleaved readout pads that share charge.

SAMPA Front-end ASIC that shapes and digitises TPC pad signals.

Central membrane High-voltage cathode at $z = 0$ of the TPC.

TPOT TPC Outer Tracker (Micromegas).

Gluckstern formula Curvature-resolution formula for N equally spaced points.

Conversion $\gamma \to e^+e^-$ in detector material.

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